

Choosing Better NLDR Layouts by Evaluating the Model in the High-dimensional Data Space

Jayani P. Gamage

Econometrics & Business Statistics, Monash University
and

Dianne Cook

Econometrics & Business Statistics, Monash University
and

Paul Harrison

MGBP, BDInstitute, Monash University
and

Michael Lydeamore

Econometrics & Business Statistics, Monash University
and

Thiyanga S. Talagala

Statistics, University of Sri Jayewardenepura

October 26, 2025

Abstract

Nonlinear dimension reduction (NLDR) techniques such as tSNE, and UMAP provide a low-dimensional representation of high-dimensional data (p - D) by applying a nonlinear transformation. NLDR often exaggerates random patterns. But NLDR views have an important role in data analysis because, if done well, they provide a concise visual (and conceptual) summary of p - D distributions. The NLDR methods and hyper-parameter choices can create wildly different representations, making it difficult to decide which is best, or whether any or all are accurate or misleading. To help assess the NLDR and decide on which, if any, is the most reasonable representation of the structure(s) present in the p - D data, we have developed an algorithm to show the 2- D NLDR model in the p - D space, viewed with a tour, a movie of linear projections. From this, one can see if the model fits everywhere, or better in some subspaces, or completely mismatches the data. Also, we can see how different methods may have similar summaries or quirks.

Keywords: high-dimensional data, dimension reduction, data visualization, statistical graphics, tour, unsupervised learning

1 Introduction

Nonlinear dimension reduction (NLDR) is popular for making a convenient low-dimensional (k - D) representation of high-dimensional (p - D) data ($k < p$). Recently developed methods include t-distributed stochastic neighbor embedding (tSNE) (Maaten & Hinton 2008), uniform manifold approximation and projection (UMAP) (McInnes et al. 2018), potential of heat-diffusion for affinity-based trajectory embedding (PHATE) algorithm (Moon et al. 2019), large-scale dimensionality reduction using triplets (TriMAP) (Amid & Warmuth 2019), and pairwise controlled manifold approximation (PaCMAP) (Wang et al. 2021).

However, the representation generated can vary dramatically from method to method, choice of hyper-parameter or even random seed, as illustrate by Figure 1. The specific method and hyper-parameters used to produce each layout (see Appendix) is not essential for the discussion. The dilemma for the analyst is which representation to use. The choice might result in different procedures used in the downstream analysis, or different inferential conclusions. Various academics have expressed concerns with current practices and procedures for choosing (e.g. Irizarry (2024), Chari & Pachter (2023)). The research described here provides new numerical and visual tools to aid with this decision.

The paper is organized as follows. Section 2 provides a summary of the literature on NLDR, and high-dimensional data visualization methods. Section 3 contains the details of the new methodology, including simulated data examples. In Section 4, we describe how to assess the best fit and identify the most accurate 2- D layout based on the proposed model diagnostics. Two applications illustrating the use of the new methodology for bioinformatics and image classification are in Section 5. Limitations and future directions are provided in

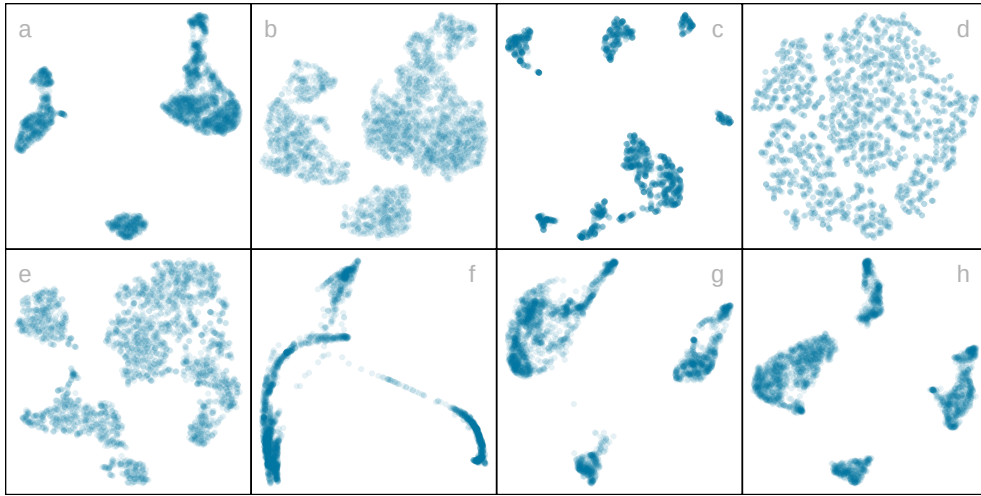


Figure 1: Eight different NLDR representations of the same data, produced by tSNE, UMAP, PHATE, TriMAP and PaCMAP with a variety of hyper-parameter choices. The variety in layouts makes it difficult to choose which best represents the data distribution.

Section 6.

2 Background

Historically, low-dimensional (k - D) representations of high-dimensional (p - D) data have been computed using multidimensional scaling (MDS) (Kruskal 1964), which includes principal components analysis (PCA) (for an overview see Jolliffe (2011)). (A contemporary comprehensive guide to MDS can be found in Borg & Groenen (2005).) The k - D representation can be considered to be a layout of points in k - D produced by an embedding procedure that maps the data from p - D . In MDS, the k - D layout is constructed by minimizing a stress function that differences distances between points in p - D with potential distances between points in k - D . Various formulations of the stress function result in non-metric scaling (Saeed et al. 2018) and isomap (Silva & Tenenbaum 2002). Challenges in working with high-dimensional data, including visualization, are outlined in Johnstone & Titterington

(2009).

Many new methods for NLDR have emerged in recent years, all designed to better capture specific structures potentially existing in p - D . Here we focus on five currently popular techniques: tSNE, UMAP, PHATE, TriMAP and PaCMAP. The methods tSNE, UMAP, TriMAP and PaCMAP can be considered for producing the k - D representation by minimizing the divergence between two inter-point distance distributions. PHATE is an example of a diffusion process spreading to capture geometric shapes, that include both global and local structure. (See [Coifman et al. \(2005\)](#) for an explanation of diffusion processes.)

The array of layouts in Figure 1 illustrate what can emerge from the choices of method and hyper-parameters, and the random seed that initiates the computation. Key structures interpreted from these views suggest: (1) highly **separated clusters** (a, b, e, g, h) with the number ranging from 3-6; (2) **stringy branches** (f), and (3) **barely separated clusters** (c, d) which would **contradict** the other representations. These contradictions arise because these methods and hyper-parameter choices provide different lenses on the interpoint distances in the data.

The alternative approach to visualizing the high-dimensional data is to use linear projections. PCA is the classical approach, resulting in a set of new variables which are linear combinations of the original variables. Tours, defined by [Asimov \(1985\)](#), broaden the scope by providing movies of linear projections, that provide views the data from all directions. (See [Lee et al. \(2021\)](#) for a review of tour methods.) There are many tour algorithms implemented, with many available in the R package `tourr` ([Wickham et al. 2011](#)), and versions enabling better interactivity in `langevitour` ([Harrison 2023](#)) and `detourr` ([Hart & Wang 2022](#)). Linear projections are a safe way to view high-dimensional data, because they do not warp the

space, so they are more faithful representations of the structure. However, linear projections can be cluttered, and global patterns can obscure local structure. The simple activity of projecting data from p - D suffers from piling (Laa et al. 2022), where data concentrates in the center of projections. NLDR is designed to escape these issues, to exaggerate structure so that it can be observed. But as a result NLDR can hallucinate wildly, to suggest patterns that are not actually present in the data.

Our proposed solution is to use the tour to examine how the NLDR is warping the space. It follows what Wickham et al. (2015) describes as *model-in-the-data-space*. The fitted model should be overlaid on the data, to examine the fit relative the spread of the observations. While this is straightforward, and commonly done when data is 2- D , it is also possible in p - D , for many models, when a tour is used.

Wickham et al. (2015) provides several examples of models overlaid on the data in p - D . In hierarchical clustering, a representation of the dendrogram using points and lines can be constructed by augmenting the data with points marking merging of clusters. Showing the movie of linear projections reveals shows how the algorithm sequentially fitted the cluster model to the data. For linear discriminant analysis or model-based clustering the model can be indicated by $(p - 1)$ - D ellipses. It is possible to see whether the elliptical shapes appropriately matches the variance of the relevant clusters, and to compare and contrast different fits. For PCA, one can display the model (a k - D plane of the reduced dimension) using wireframes of transformed cubes. Using a wireframe is the approach we take here, to represent the NLDR model in p - D .

3 Method

3.1 What is the NLDR model?

At first glance, thinking of NLDR as a modeling technique might seem strange. It is a simplified representation or abstraction of a system, process, or phenomenon in the real world. The p - D observations are the realization of the phenomenon, and the k - D NLDR layout is the simplified representation. Typically, $k = 2$, which is used for the rest of this paper. From a statistical perspective we can consider the distances between points in the 2- D layout to be variance that the model explains, and the (relative) difference with their distances in p - D is the error, or unexplained variance. We can also imagine that the positioning of points in 2- D represent the fitted values, that will have some prescribed position in p - D that can be compared with their observed values. This is the conceptual framework underlying the more formal versions of factor analysis ([Jöreskog 1969](#)) and MDS. (Note that, for this thinking the full p - D data needs to be available, not just the interpoint distances.)

We define the NLDR as a function $g: \mathbb{R}^{n \times p} \rightarrow \mathbb{R}^{n \times 2}$, with hyper-parameters θ . These parameters, θ , depend on the choice of g , and can be considered part of model fitting in the traditional sense. Common choices for g include functions used in tSNE, UMAP, PHATE, TriMAP, PaCMAP, or MDS, although in theory any function that does this mapping is suitable.

With our goal being to make a representation of this 2- D layout that can be lifted into high-dimensional space, the layout needs to be augmented to include neighbor information. A simple approach would be to triangulate the points and add edges. A more stable

approach is to first bin the data, reducing it from n to $m \leq n$ observations, and connect the bin centroids. We recommend using a hexagon grid because it better reflects the data distribution and has less artifacts than a rectangular grid. This process serves to reduce some noisiness in the resulting surface shown in p - D . The steps in this process are shown in Figure 2, and documented below.

To illustrate the method and how to use it to choose a reasonable layout, we use 7- D simulated data, which we call the “2NC7” data. It has two separated nonlinear clusters, one forming a 2- D curved shape, and the other a 3- D curved shape, each consisting of 1000 observations. The first four variables hold this cluster structure, and the remaining three are purely noise. We would consider $T = (X_1, X_2, X_3, X_4)$ to be the geometric structure (true model) that we hope to capture. This data is sufficiently simple, with just two complexities (two separated curvilinear clusters, and two different implicit dimensions), to adequately explain the new method. The applications section contains two practical examples where NLDR has been used in published work.

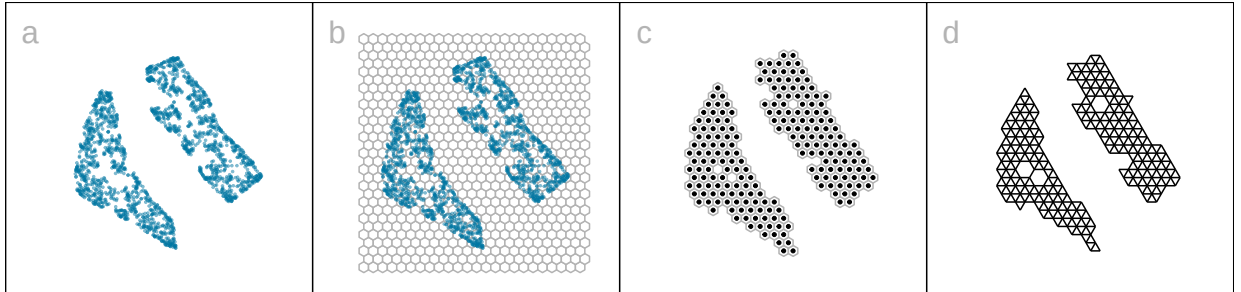


Figure 2: Key steps for constructing the model on the tSNE layout ($k = 2$) of 2NC7: (a) data, (b) hexagon bins, (c) bin centroids, and (d) triangulated centroids. The 2NC7 data is shown.

3.2 Algorithm to represent the model in 2-D

3.2.1 Scale the data

Because we are working with distances between points, starting with data having a standard scale, e.g. $[0, 1]$, is recommended. The default should take the aspect ratio produced by the NLDR $(r_1, r_2, \dots, r_k)$ into account. When $k = 2$, as in hexagon binning, the default range is $[0, y_{i,\max}]$, $i = 1, 2$, where $y_{1,\max} = 1$ and $y_{2,\max} = r_2/r_1$ (Figure 2). If the NLDR aspect ratio is ignored then set $y_{2,\max} = 1$.

3.2.2 Hexagon grid configuration

Although there are several implementations of hexagon binning (Carr et al. 1987), and a published paper (Carr et al. 2023), surprisingly, none has sufficient detail or components that produce everything needed for this project. So we described the process used here. Figure 3 illustrates the notation used.

The 2-D hexagon grid is defined by its bin centroids. Each hexagon, H_h ($h = 1, \dots, b$) is uniquely described by centroid, $C_h^{(2)} = (c_{h1}, c_{h2})$. The number of bins in each direction is denoted as (b_1, b_2) , with $b = b_1 \times b_2$ being the total number of bins. We expect the user to provide just b_1 and we calculate b_2 using the NLDR ratio, to compute the grid.

To ensure that the grid covers the range of data values a buffer parameter (q) is set as a proportion of the range. By default, $q = 0.1$. The buffer should be extending a full hexagon width (a_1) and height (a_2) beyond the data, in all directions. The lower left position where the grid starts is defined as (s_1, s_2) , and corresponds to the centroid of the lowest left hexagon, $C_1^{(2)} = (c_{11}, c_{12})$. This must be smaller than the minimum data value. Because it

is one buffer unit, q below the minimum data values, $s_1 = -q$ and $s_2 = -qr_2$.

The value for b_2 is computed by fixing b_1 . Considering the upper bound of the first NLDR component, $a_1 > (1 + 2q)/(b_1 - 1)$. Similarly, for the second NLDR component,

$$a_2 \geq \frac{r_2 + q(1 + r_2)}{(b_2 - 1)}.$$

Since $a_2 = \sqrt{3}a_1/2$ for regular hexagons,

$$a_1 \geq \frac{2[r_2 + q(1 + r_2)]}{\sqrt{3}(b_2 - 1)}.$$

This is a linear optimization problem. Therefore, the optimal solution must occur on a vertex. Therefore,

$$b_2 = \left\lceil 1 + \frac{2[r_2 + q(1 + r_2)](b_1 - 1)}{\sqrt{3}(1 + 2q)} \right\rceil. \quad (1)$$

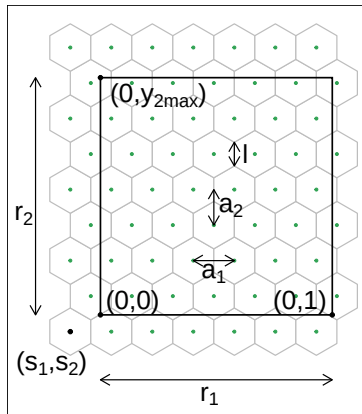


Figure 3: The components of the hexagon grid illustrating notation.

3.2.3 Binning the data

Observations are grouped into bins based on their nearest centroid. This produces a reduction in size of the data from n to m , where $m \leq b$ (total number of bins). This can be defined using the function $u : \mathbb{R}^{n \times 2} \rightarrow \mathbb{R}^{m \times 2}$, where

$$u(i) = \arg \min_{j=1, \dots, b} \sqrt{(y_{i1} - C_{j1}^{(2)})^2 + (y_{i2} - C_{j2}^{(2)})^2},$$

maps observation i into $H_h = \{i | u(i) = h\}$.

By default, the bin centroid is used for describing a hexagon (as done in Figure 2 (c)), but any measure of center, such as a mean or weighted mean of the points within each hexagon, could be used. The bin centers, and the binned data, are the two important components needed to render the model representation in high dimensions.

3.2.4 Indicating neighborhood

Delaunay triangulation (Lee & Schachter 1980, Gebhardt et al. 2024) is used to connect points so that edges indicate neighboring observations, in both the NLDR layout (Figure 2 (d)) and the p - D model representation. When the data has been binned the triangulation connects centroids. The edges preserve the neighborhood information from the 2- D representation when the model is lifted into p - D .

3.3 Rendering the model in p -D

The last step is to lift the 2- D model into p - D by computing p - D vectors that represent bin centroids. We use the p - D mean of the points in a given hexagon, H_h , denoted $C_h^{(p)}$, to map the centroid $C_h^{(2)} = (c_{h1}, c_{h2})$ to a point in p - D . Let the j^{th} component of the p - D mean be

$$C_{hj}^{(p)} = \frac{1}{n_h} \sum_{i=1}^{n_h} x_{hij}, \quad h = 1, \dots, b; \quad j = 1, \dots, p; \quad n_h > 0.$$

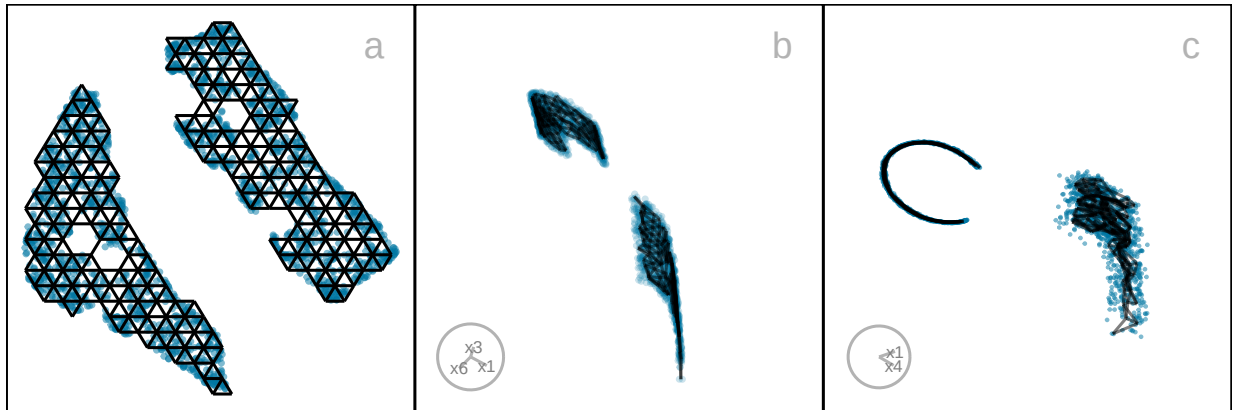


Figure 4: Lifting the 2- D fitted model into p - D . Two projections of the p - D fitted model overlaying the data are shown in b, c. The fit is reasonably tight with the data in one cluster (top one in b), but slightly less so in the other cluster probably because it is 3- D . Notice also that, in the 2- D layout the two clusters have internal gaps which creates a model with some holes. This lacy pattern happens regardless of the hyper-parameter choice, but this doesn't severely impact the p - D model representation.

3.4 Measuring the fit

All NLDR methods internally optimize a quantity to produce a layout for any particular hyper-parameter set. These are not always made available in the model output, and may not be universally comparable between hyper-parameter choices and methods.

Several common metrics are often used to assess the quality of any NLDR layout. The RNX

curve quantifies the neighborhood agreement between p - D and k - D spaces, by computing the area under the curve ($ARNX$) across a range of neighborhood scales (Lee et al. 2015). Random Triplet Accuracy (RTA) and Centroid Triplet Accuracy quantify global structure preservation by comparing relative distances among sampled points or class centroids (Wang et al. 2021). The Shepard diagram (Shepard 1962) and its associated Spearman correlation (SC) (Spearman 1961) between p - D and k - D distances. The Global Score (GS) measures how well an embedding retains the overall geometry of the data relative to a PCA baseline (Amid & Warmuth 2019). KNN Accuracy evaluates neighborhood consistency using leave-one-out classification, and SVM Accuracy assesses neighborhood cohesion with a nonlinear decision boundary (Wang et al. 2021). Because these two require a response variable, they can't be applied for our purposes.

None of the above measures is particularly well-suited to assessing our model fit, as we will show later. Thus, we need a different approach to measuring model fit. Because the model here is similar to a confirmatory factor analysis model (see a general explanation in Brown (2015)), $\hat{T} + E$, our approach is similar to the ones used in this area. It is based on “residuals” computed as the difference between the fitted model and observed values in p - D . Observations are associated with their bin center, $C_h^{(p)}$, which are also considered to be the *fitted values*. In factor analysis language, these fitted values might also be denoted as $\hat{X}$. The error is computed by taking the squared p - D Euclidean distance, of points from their bin centroid, which we will call hexbin error (HBE):

$$HBE = \sqrt{\frac{1}{n} \sum_{h=1}^m \sum_{i=1}^{n_h} \sum_{j=1}^p (\mathbf{x}_{hij} - C_{hj}^{(p)})^2} \quad (2)$$

where n is the number of observations, m is the number of non-empty bins, n_h is the number of observations in h^{th} bin, p is the number of variables and $\mathbf{x}_{hij}$ is the j^{th} dimensional data of i^{th} observation in h^{th} hexagon. We can consider $e_{hi} = \sqrt{\sum_{j=1}^p (\mathbf{x}_{hij} - C_{hj}^{(p)})^2}$ to be the residual for each observation. Figure 5 shows plots of e as a density (a), coloring the points in the NLDR layout (b) and the points in a tour (c). It can be seen that the biggest residuals are in one cluster, which occurs due to the intentional design that one cluster is slightly 3- D and perfectly captured by a 2- D layout.

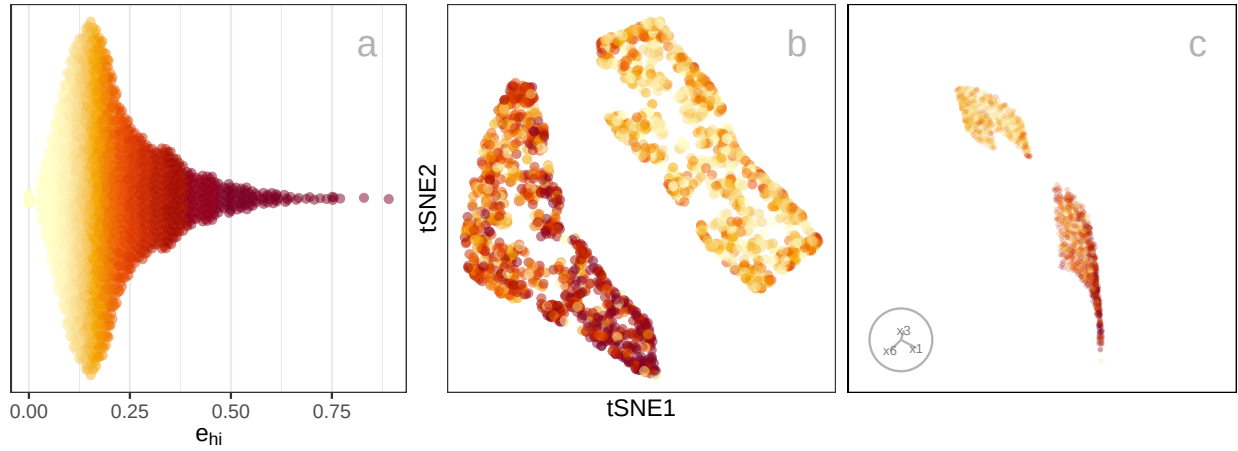


Figure 5: Examining the distribution of residuals in a jittered dotplot (a), 2- D NLDR layout (b) and a tour of 4- D data space (c). Color indicates residual (e_{hi}), dark color indicating high value. Most large residuals are distributed in one cluster (bottom one in c) and most small residuals are distributed in the other cluster.

3.5 Prediction into 2- D

NLDR methods are primarily designed for visualization and exploration rather than reconstruction, and do not explicitly provide out-of-sample prediction. Of the five methods studied here, only UMAP provides a `predict()` function for embedding new data points based on the learned manifold (Konopka 2023). Several other approaches, not used here, PCA, neural network autoencoders (Hinton & Salakhutdinov 2006) and parametric tSNE

([van der Maaten 2009](#)) support reverse mapping.

A benefit of our approach is that for any NLDR method, it provides a way to predict the layout position of a new observation, x' . The steps are (1) determine the closest bin centroid in p -D, $C_h^{(p)}$ and (2) predict the embedding to be the bin centroid in 2-D, $C_h^{(2)}$.

3.6 Tuning

The model fitting is based on these parameters:

- hexagon bin parameters
 - bottom left bin position (s_1, s_2),
 - the number of bins in the horizontal direction (b_1), which controls the number of bins the vertical direction (b_2), total number of bins (b), and total number of non-empty bins (m).
- low count bin removal using a bin density cutoff, based on standardised bin counts ($w_h = n_h/n$, $h = 1, \dots m$). (Note that, density can be also computed and used, as $d_h = w_h/A$ where A is the area of the hexagon.)

Default values are provided for each of these, but deciding on the best model fit is assisted by examining a range of values. The default number of bins $b = b_1 \times b_2$ is computed based on the sample size, by setting $b_1 = n^{1/3}$, consistent with the Diaconis-Freedman rule ([Freedman & Diaconis 1981](#)). The value of b_2 is determined analytically by b_1, q, r_2 (Equation 1). Values of b_1 between 2 and $b_1 = \sqrt{n/r_2}$ are recommended, where the dependence on r_2 reflects the preservation of aspect ratio in the NLDR layout.

Figure 6 shows the hexbin grids for three choices of b_1 . While the number of bins is the common parameter to modify, bin start positions (s_1, s_2) can be worth experimenting with also because it can also change bin counts.

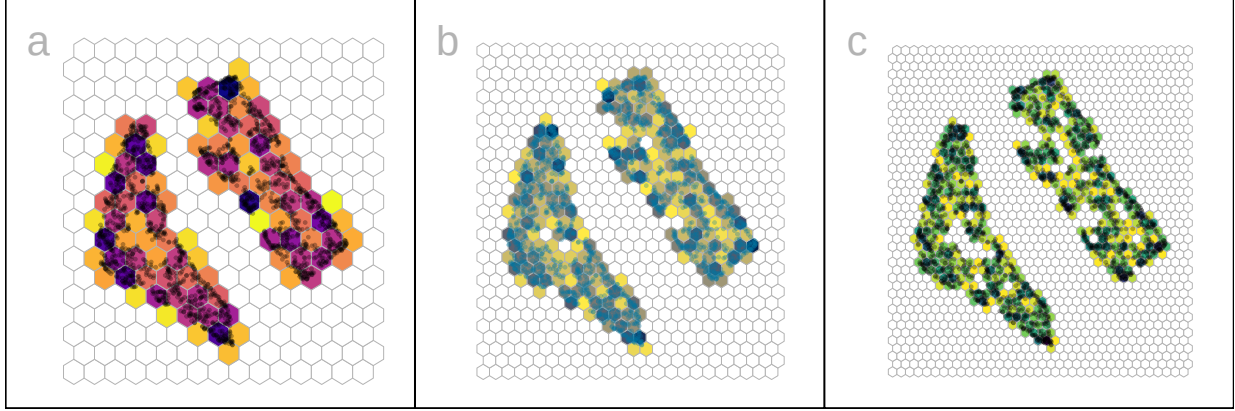


Figure 6: Hexbin density plots of tSNE layout of the 2NC7 data, using three different b_1 specifications yielding different b_2, b, m : (a) **15**, 18, 270, 98, (b) **24**, 29, 696, 209, and (c) **35**, 42, 1470, 386. Color indicates standardized counts, dark indicating high count and light indicates low count. At the smallest bin size, the data structure is discontinuous, suggesting that there are too many bins.

It is worthwhile to consider what are desirable aspects of a hexbin result, that maps to summarising the p - D fit well. The binning should capture the underlying data distribution closely, with minimum number of necessary bins. An ideal binning might be indicated by a more uniform distribution of bin counts, or having few relatively empty bins. To help with this assessment average standardized bin count ($\bar{w} = \sum w_h/m$), average density ($\bar{d} = \sum d_h/m$) and proportion of non-empty bins (m/b), are also computed. Figure 7 shows some choices of plots of these quantities for a single NLDR layout, with three choices of a_1 indicated. Some expectations and reasoning for these plots is:

- HBE will increase as a_1 increases, so good choices will be just before a big increase.

In plot a, HBE changes fairly steadily so there is no easy choice to make.

- HBE can also be examined against average standardised bin count or average density (plot b). This is similar to the comparison with a_1 but to use when comparing different NLDR layouts. Different layouts might produce different density of points, which will not be captured well by a comparison of HBE vs a_1 .
- The proportion of non-empty bins is interesting to examine across different binwidths (plot c). A good binning should have just the right amount of bins to neatly cover the shape of the data, and no more or less. As binwidth gets smaller, m/b should roughly get bigger.
- Bins with a small number of observations might be removed to sharpen the wireframe model. This can have adverse effects, though - failing to extend the wireframe into sparse areas, or resulting in holes in the wireframe. Plot d shows the relationship between HBE (computed for all observations despite some bin removal) and the density cutoff used to remove bins. For all three chosen bin widths a small number of bins can be removed without affecting HBE.

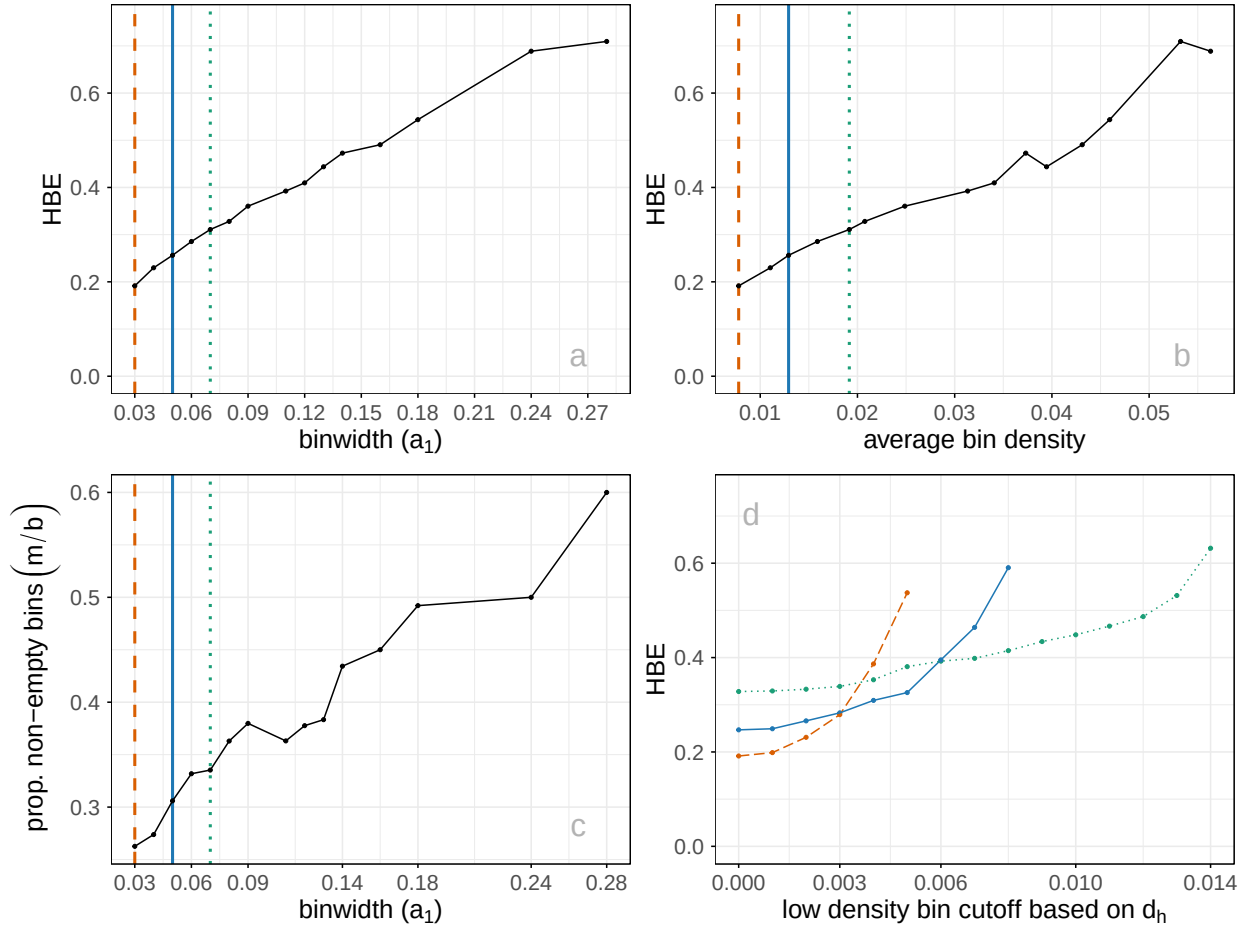


Figure 7: Various plots to help tune the model fit: (a) HBE vs a_1 , (b) HBE vs $\bar{d}_h$, (c) proportion of non-empty bins vs a_1 , (d) HBE vs w_h cutoff used for removing low count bins. Color indicates the three binwidths show in Figure 6: 0.03 (orange dashed), 0.05 (blue solid), and 0.07 (green dotted). The better model fit will have low HBE, but reasonably sized bins that capture the data sufficiently. Proportion of non-empty bins tends to increase with a_1 (c). Removing a few low count bins doesn't substantially change HBE, for all three binwidths (d).

3.7 Interactive graphics

Matching points in the 2- D layout with their positions in p - D is useful when tuning the fit.

This can be used to examine the fitted model in some subspaces in p - D , in particular in association with residual plots.

The 2- D layout and the langevitour view with the fitted model overlaid can be linked using

a **browsable HTML widget**. A rectangular “brush” is used to select points in one plot, which will highlight the corresponding points in the other plot(s). Because the langevitour is dynamic, brush events that become active will pause the animation, so that a user can interrogate the current view. This approach will be illustrated on the examples, to show how it can help to understand how the NLDR has organised the observations, and learn where it does not do well.

4 Choosing the best 2-*D* layout

Figure 8 illustrates the approach to compare the fits for different representations and assess the strength of any fit. What does it mean to be a best fit for this problem? Analysts use an NLDR layout to display the structure present in high-dimensional data in a convenient 2-*D* display. It is a competitor to linear dimension reduction that can better represent nonlinear associations such as clusters. However, these methods can hallucinate, suggesting patterns that don’t exist, and grossly exaggerate other patterns. Having a layout that best fits the high-dimensional structure is desirable but more important is to identify bad representations so they can be avoided. The goal is to help users decide on a the most useful and appropriate low-dimensional representation of the high-dimensional data.

A particular pattern that we commonly see is that analysts tend to pick layouts with clusters that have big separations between them. When you examine their data in a tour, it is almost always that we see there are no big separations, and actually often the suggested clusters are not even present. While we don’t expect that analysts include animated gifs of tours in their papers, we should expect that any 2-*D* representation adequately indicates the

clustering that is present, and honestly show lack of separation or lack of clustering when it doesn't exist. It is important for analysts to have tools to select the accurate representation not the pretty but wrong representation.

To decide on a layout an analyst needs:

- a selection of NLDR representations made with a range of hyper-parameter choices and possibly different methods (tSNE, UMAP, ...).
- a range of model fits made by varying bin size and low density bin removal.
- calculated HBE for each layout, when it is transformed into the p - D space.
- and the ability to examine the fit in the data space.

Comparing the HBE to obtain the best fit is appropriate if the same NLDR method is used. However, because the HBE is computed on p - D data it measures the fit between model and data so it can also be used to compare the fit of different NLDR methods. A lower HBE indicates a better NLDR representation.

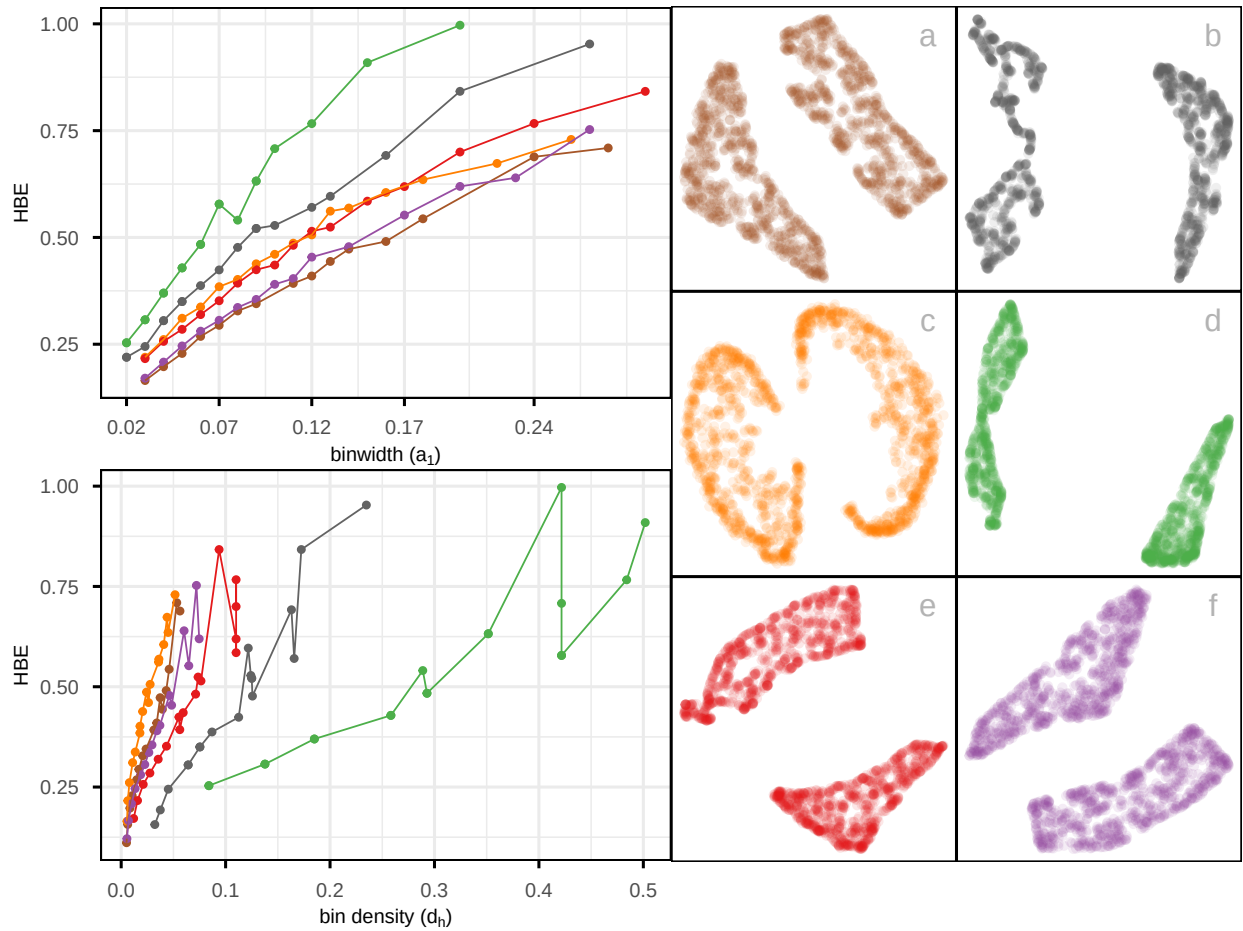


Figure 8: Assessing which of the 6 NLDR layouts on the 2NC7 data is the better representation using HBE for varying binwidth (a_1). Color used for the lines and points in the left plot and in the scatterplots represents NLDR layout (a-f). Layout d is universally poor. Layouts a, b, e that show two close clusters are universally suboptimal. Layout b with little separation performs well at tiny binwidth (where most points are in their own bin) and poorly as binwidth increases. Layout e has small separation with oddly shaped clusters. Layout a is the best choice.

Figure 9 compares the metrics ARNX, RTA, SC, GS, along with HBE computed on $a_1 = 0.05$ for the six layouts shown in Figure 8. This is a parallel coordinate plot where the y-axis shows a normalized score to ensure the metrics are on the same scale. Each line corresponds to one layout. The metric ARNX has been reversed so that it aligns with HBE - the lower the value the better the layout.

There is some agreement between the metrics. All, except ARNX agree that layout d is

worst. All agree that layout f is best or very close to best. Layout a is best according to HBE and ARNX but considered to be much less optimal by RTA, SC and GS. Layout c is considered favorably by RTA, SC and GS but to be very poor by ARNX. This illustrates how difficult it is to use the numerical metrics alone to decide on the best layout.

Ironically, RTA, GS and SC should be interpreted as the higher the value the better the layout, but here they agree with HBE and ARNX when used in reverse! The problem with SC is that correlation is not a good measure in the presence of clusters - the further the clusters are apart in the layout produces a Shepard plot with two clusters of distances which will produce a high correlation value. Similar reasoning may explain why RTA and GS behave similarly. They may perform better when the structure being summarised is a single cluster of data lying on a nonlinear manifold.

When the metrics disagree it causes confusion for the analyst, and thus provides a temptation to choose the nicest looking layout, even though it may be sub-optimal. Because HBE is accompanied with a representation of the layout in p - D to compare with the observed data, it can help to add more clarity in making decisions.

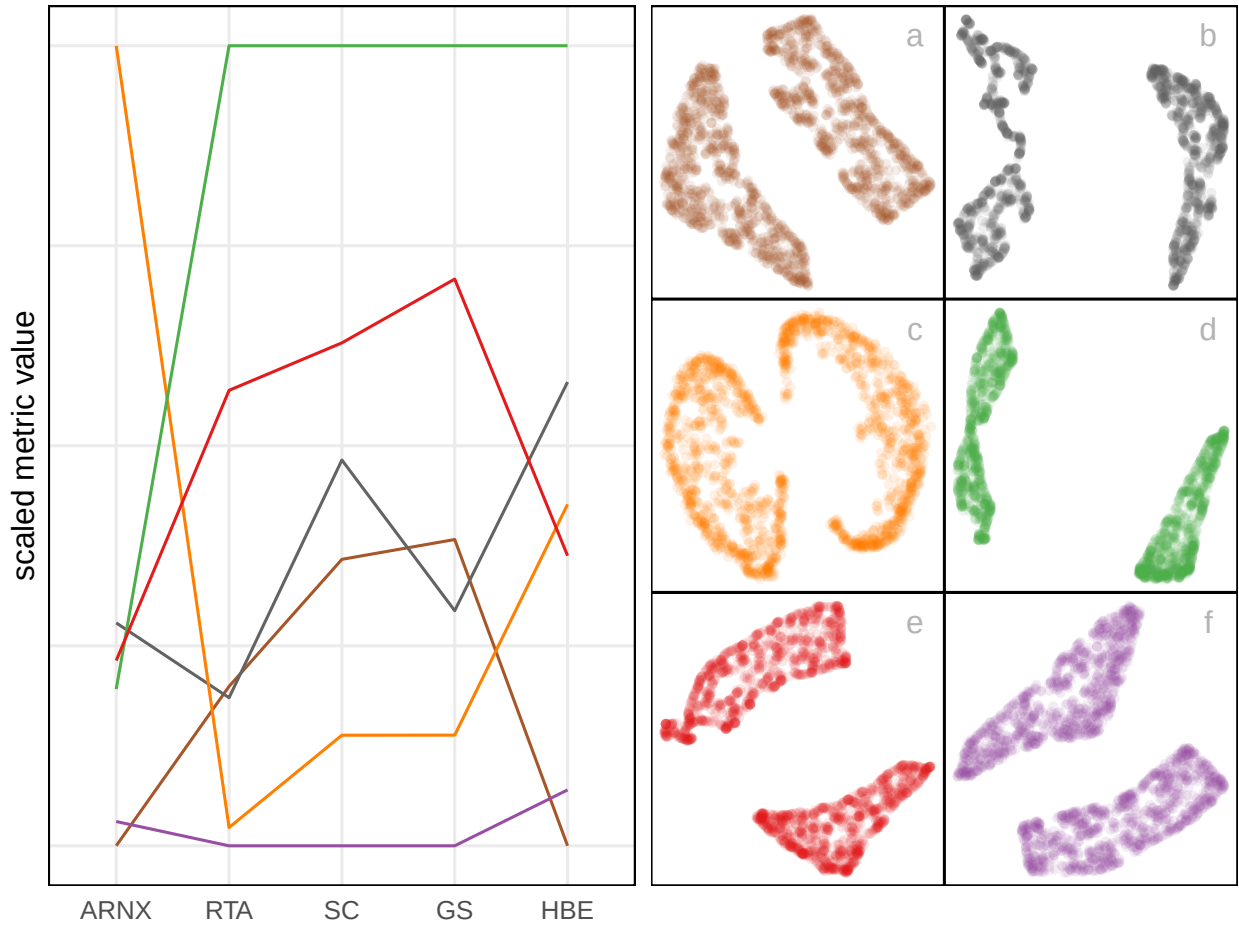


Figure 9: Comparison of scaled evaluation metrics (RTA, ARNX, SC, GS, and HBE using $a_1 = 0.05$) for the six NLDR layouts computed on the 2NC7 data using a parallel coordinate plot. Color of the line indicates NLDR layout. ARNX and RGS are reversed so that lower is best like the other metrics. There is some general agreement, and small disagreements, with ARNX being most different. All agree that layout f is good, and almost all that d is bad. Interestingly, there is disagreement on layout a, which is almost the same as layout f.

5 Applications

To illustrate the approach we use two examples: PBMC3k data (single cell gene expression) where an NLDR layout is used to represent cluster structure present in the p - D data, and MNIST hand-written digits where NLDR is used to represent essentially a low-dimensional nonlinear manifold in p - D .

5.1 PBMC3k

This is a benchmark single-cell RNA-Seq data set collected on Human Peripheral Blood Mononuclear Cells (PBMC3k) as used in [10x Genomics \(2016\)](#). Single-cell data measures the gene expression of individual cells in a sample of tissue (see for example, [Haque et al. \(2017\)](#)). This type of data is used to obtain an understanding of cellular level behavior and heterogeneity in their activity. Clustering of single-cell data is used to identify groups of cells with similar expression profiles. NLDR is often used to summarize the cluster structure. Usually, NLDR does not use the cluster labels to compute the layout, but uses color to represent the cluster labels when it is plotted.

In this data there are 2622 single cells and 1000 gene expressions (variables). Following the same pre-processing as [Chen et al. \(2024\)](#), different NLDR techniques were performed on the first nine principal components. Figure 1 shows this data using a variety of methods, and different hyper-parameters. You can see that the result is wildly different depending on the choices. Layout a is a reproduction of the layout that was published in [Chen et al. \(2024\)](#). This layout suggests that the data has three very well separated clusters, each with an odd shape. The question is whether this accurately represents the cluster structure in the data, or whether they should have chosen b or c or d or e or f or g or h. This is what our new method can help with – to decide which is the more accurate $2-D$ representation of the cluster structure in the $p-D$ data.

Figure 10 shows HBE across a range of binwidths (a_1) for each of the layouts in Figure 1. The layouts were generated using tSNE and UMAP with various hyper-parameter settings, while PHATE, PaCMAP, and TriMAP were applied using their default settings. Lines are

color coded to match the color of the layouts shown on the right. Lower HBE indicates the better fit. Using a range of binwidths shows how the model changes, with possibly the best model being one that is universally low HBE across all binwidths. It can be seen that layout f is sub-optimal with universally higher HBE. Layout a, the published one, is better but it is not as good as layouts b, d, or e. With some imagination layout d perhaps shows three barely distinguishable clusters. Layout e shows three, possibly four, clusters that are more separated. The choice reduces from eight to these two. Layout d has slightly better HBE when the a_1 is small, but layout e beats it at larger values. Thus we could argue that layout e is the most accurate representation of the cluster structure, of these eight.

To further assess the choices, we need to look at the model in the data space, by using a tour to show the wireframe model overlaid on the data in the 9- D space (Figure 11). Here we compare the published layout (a) versus what we argue is the best layout (e). The top row (a1, a2, a3) correspond to the published layout and the bottom row (e1, e2, e3) correspond to the optimal choice according to our procedure. The middle and right plots show two projections. The primary difference between the two models is that the model of layout e does not fill out to the extent of the data but concentrates in the center of each point cloud. Both suggest that three clusters is a reasonable interpretation of the structure, but layout e more accurately reflects the separation between them, which is small.

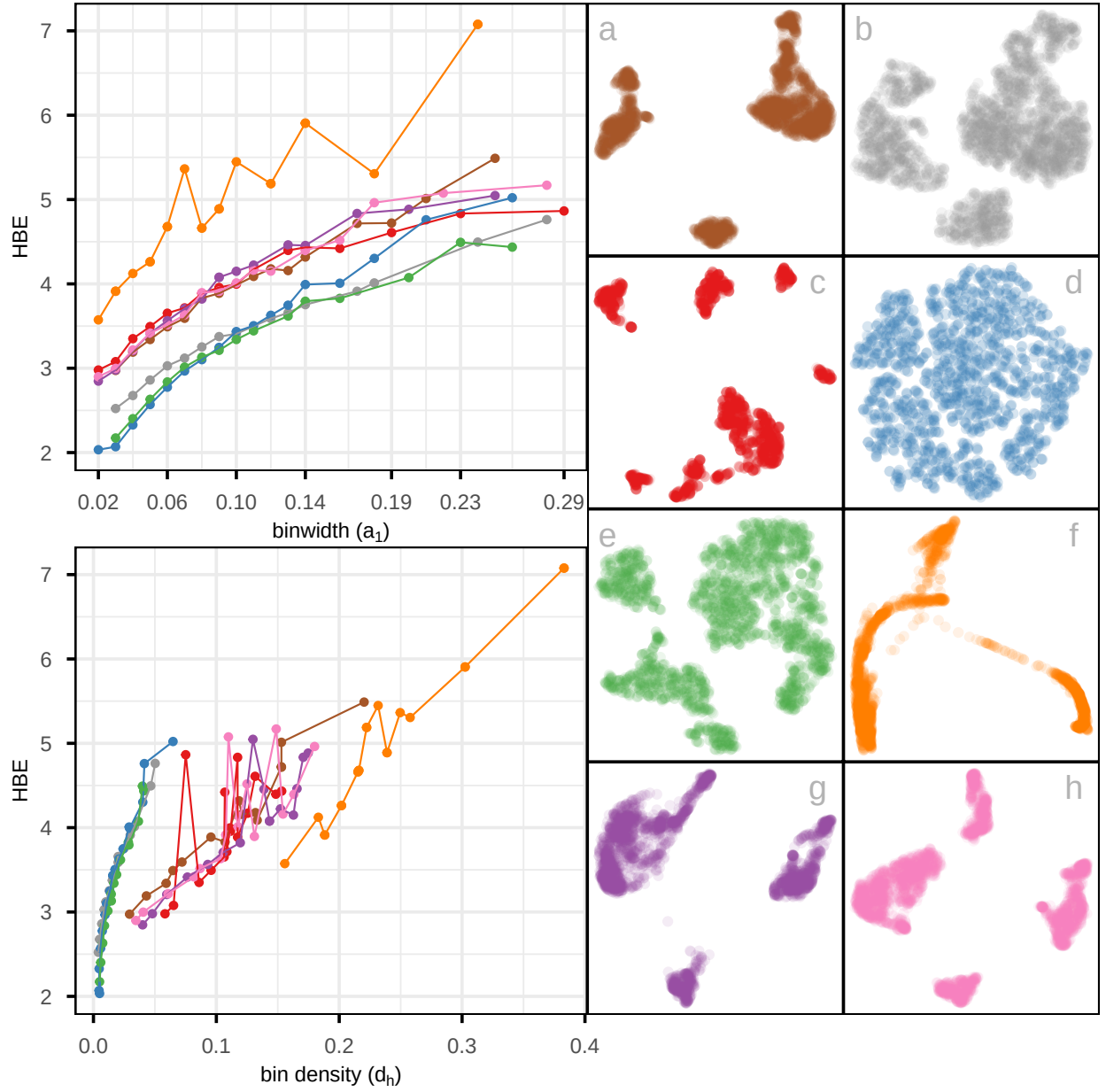


Figure 10: Assessing which of the 8 NLDR layouts on the PBMC3k data (shown in Figure 1) is the better representation using HBE for varying binwidth (a_1). Color used for the lines and points in the left plot and in the scatterplots represents NLDR layout (a-h). Layout f is universally poor. Layouts a, c, g, h that show large separations between clusters are universally suboptimal. Layout d with little separation performs well at tiny binwidth (where most points are in their own bin) and poorly as binwidth increases. The choice of best is between layouts b and e, that have small separations between oddly shaped clusters. Layout e is chosen as the best.

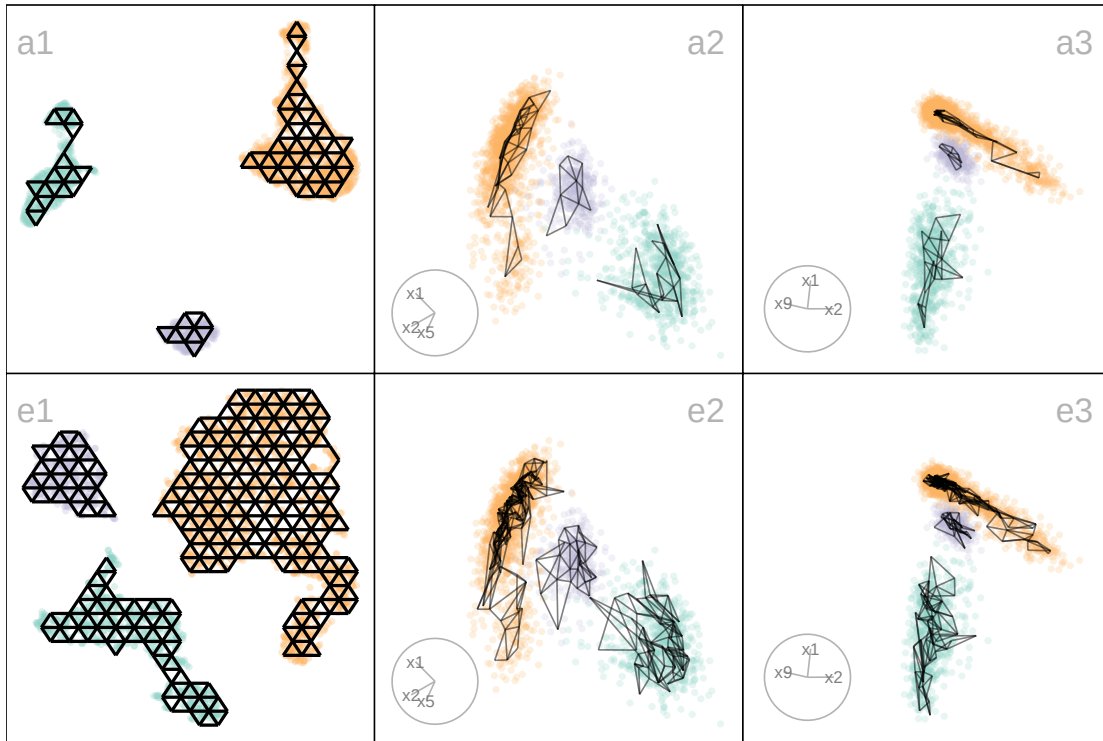


Figure 11: Compare the published 2- D layout (a) made with UMAP and the 2- D layout selected by HBE plot (e) made by tSNE. The two plots on the right show projections from a tour, with the models overlaid. The published layout a suggested three very separated clusters, but this is not present in the data. While there may be three clusters they are not well-separated. The difference in model fit also indicates this: the published layout a does not spread out fully into the point cloud like the model generated from layout e. This supports the choice that layout e is the better representation of the data, because it does not exaggerate separation between clusters.

5.2 MNIST hand-written digits

The digit “1” of the MNIST dataset ([LeCun et al. 1998](#)) consists of 7877 grayscale images of handwritten “1”s. Each image is 28×28 pixels which corresponds to 784 variables. The first 10 principal components, explaining 83% of the total variation, are used. This data essentially lies on a nonlinear manifold in the high dimensions, defined by the shapes that “1”s make when sketched. We expect that the best layout captures this type of structure and does not exhibit distinct clusters.

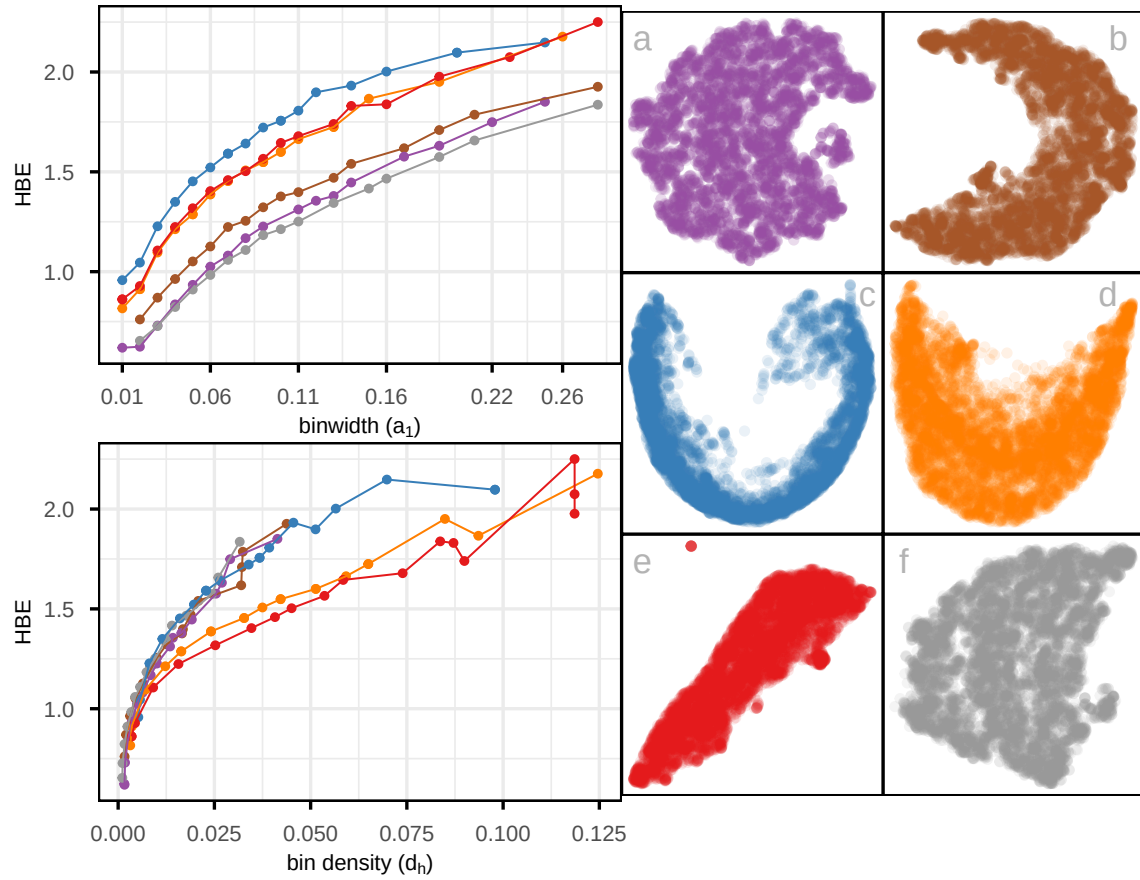


Figure 12: Assessing which of the 6 NLDR layouts of the MNIST digit 1 data is the better representation using HBE for varying binwidth (a_1). Colour is used for the lines and points in the left plot to match the scatterplots of the NLDR layouts (a-f). Layout c is universally poor. Layouts a, f that show a big cluster and a small circular cluster are universally optimal. Layout a performs well at tiny binwidth (where most points are in their own bin) and not as well as f with larger binwidth, thus layout f is the best choice.

Figure 12 compares the fit of six layouts computed using UMAP (b), PHATE (c), TriMAP (d), PaCMAP (e) with default hyper-parameter setting and two tSNE runs, one with default hyper-parameter setting (a) and the other changing perplexity to 89 (f). The layouts are reasonably similar in that they all have the observations in a single blob. Some (b, c) have a more curved shape than others. Layout e is the most different having a linear shape, and a single very large outlier. Both a and f have a small clump of points perhaps slightly disconnected from the other points, in the lower to middle right.

The layout plots are colored to match the lines in the HBE vs binwidth (a_1) plot. Layouts a, b and f fit the data better than c, d, e, and layout f appears to be the best fit. Figure 13 shows this model in the data space in two projections from a tour. The data is curved in the 10- D space, and the fitted model captures this curve. The small clump of points in the 2- D layout is highlighted in both displays. These are almost all inside the curve of the bulk of points and are sparsely located. The fact that they are packed together in the 2- D layout is likely due to the handling of density differences by the NLDR.

The next step is to investigate the 2- D layout to understand what information is learned from this representation. Figure 14 summarizes this investigation. Plot a shows the layout with points colored by their residual value - darker color indicates larger residual and poor fit. The plots b, c, d, e show samples of hand-written digits taken from inside the colored boxes. Going from top to bottom around the curve shape we can see that the “1”s are drawn with from right slant to a left slant. The “1”s in d (black box) tend to have the extra up stroke but are quite varied in appearance. The “1”s shown in the plots labelled e correspond to points with big residuals. They can be seen to be more strangely drawn than the others. Overall, this 2- D layout shows a useful way to summarize the variation in way “1”s are drawn.

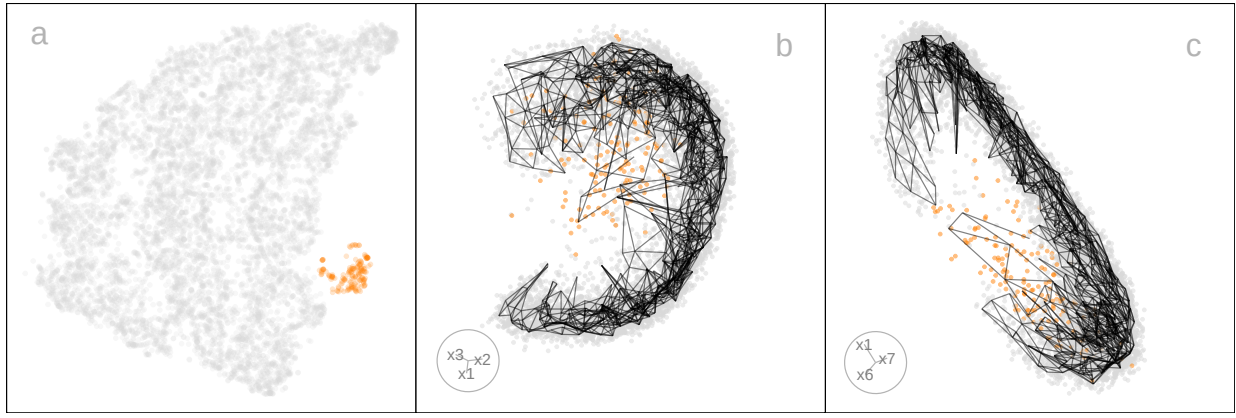


Figure 13: The tSNE layout of the MNIST digit 1 data shows a big nonlinear cluster (grey) and a small cluster (orange) located very close to the one corner of the big cluster in 2- D (a). The MNIST digit 1 data has a nonlinear structure in 10- D . Two 2- D projections from a tour on 10- D reveal that the closeness of the clusters in 10- D and the twisted pattern of the model fit with tSNE. The brushing feature in the linked plots helps in visualizing the closeness of the small cluster to the big cluster.

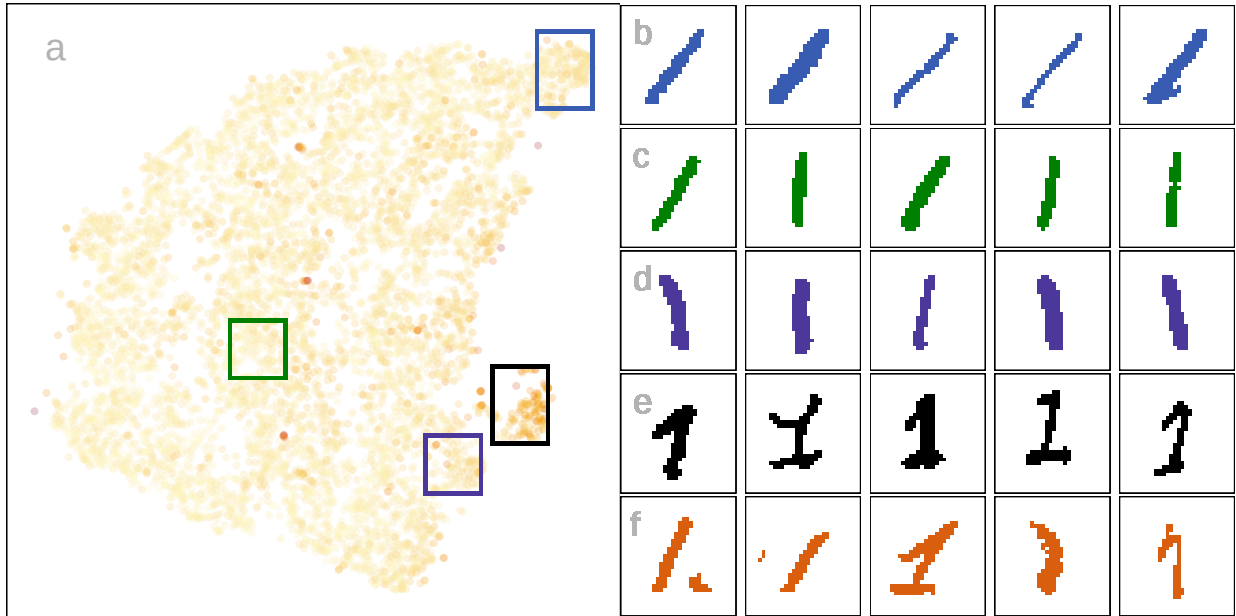


Figure 14: There is a pattern to error of the fit of the model for 2- D layout of the MNIST digit 1 data. On the left is the NLDR layout (a), and at right (b-e) are images of samples of observations taken at locations along the big cluster, and the small cluster, showing how the 1's were drawn. Set (f) are images corresponding to large residuals in the big cluster. Along the big cluster, the angle of digit 1 changes (b-d). The small cluster has larger residuals, and the images show that these tend to be European style with a flag at the top, and a base at the bottom. The set in (f) show various poorly written digits.

6 Discussion

We have developed an approach to help assess and compare NLDR layouts, generated by different methods and hyper-parameter choice(s). It depends on conceptualizing the 2- D layout as a model, allowing for the creation of a wireframe representation of the model that can be lifted into p - D . The fit is assessed by viewing the model in the data space, computing residuals and HBE. Different layouts can be compared using the HBE, and provides quantitative and objective methods for deciding on the most suitable NLDR layout to represent the p - D data. It also provides a way to predict the values of new p - D observations in the 2- D , which could be useful for implementing uncertainty checks such as using training and testing samples.

This work can be applied with existing metrics for evaluating NLDR layout, such as ARNX, RTA, sc, and RGS. It provides an additional evaluation metric, and allows any layout to be viewed in the p - D data space. This latter aspect can help to disentangle conflicting suggestions by the different metrics.

Two examples illustrating usage are provided: the PBMC3k data where the NLDR is summarizing clustering in p - D and hand-written digits illustrating how NLDR represents an intrinsically lower dimensional nonlinear manifold. We examined a typical published usage of UMAP with the PBMC3k dataset (Chen, 2024). As is typical of UMAP layout with default settings, the separation between clusters is grossly exaggerated. The layout even suggests separation where there is none. Our approach provides a way to objectively choose the layout and hopefully avoids the use of misleading layouts in the future. In the hand-written digits we illustrate how our model fit statistics show that a flat disc layout is

superior to the curved shaped layouts, and how to identify oddly written “1”s using the residuals of the fitted model.

Additional exploration of metrics to summarize the fit could be a new direction for the work. The difficulty is capturing nonlinear fits, for which Euclidean distance can be sub-optimal. We have used a very simple approach based on clustering methods, Euclidean distances to nearest centroid, which can approximate nonlinear patterns. Other cluster metrics would be natural choices to explore.

This new method also reveals some interesting curiosities about NLDR procedures. The fitted model appears as a “pancake” in some data where clusters are regularly shaped and high-dimensional, for some methods but not others, which is odd. One can imagine that if algorithms are initiated using principal components then some ordering of points along the major axes might generate this pattern. Alternatively, if local distances dominate the algorithm then it might be possible to see this pattern with well-separated regular clusters. We also demonstrated that there is a tendency for NLDR algorithms to be confused by different density in the data space, and some patterns in the layout are due to density differences rather than nonlinear associations between variables.

Most NLDR methods only provide a 2- D but if a k - D ($k > 2$) layout is provided the approach developed here could be extended. Binning into cubes could be done in 3- D or higher, relatively easily, and used as the basis for a wireframe of the fitted model. [Barber et al. \(1996\)](#) (and the associated software [Laurent \(2023\)](#)) has an algorithm for a convex hull in p - D which serves as an inspiration. A simpler approach using k -means clustering to provide centroids could also be possible, but the complication would be to determine how to connect the centroids into an appropriate wireframe.

XXX Better tools to compare two layouts in p - D would be nice.

The new methodology is accompanied by an R package called `quollr`, so that it is readily usable and broadly accessible. The package has methods to fit the model, compute diagnostics and also visualize the results, with interactivity. We have primarily used the `langevitour` software (Harrison 2023) to view the model in the data space, but other tour software such as `tourr` (Wickham et al. 2011) and `detourr` (Hart & Wang 2022) could be also used.

7 Supplementary Materials

All the materials to reproduce the paper can be found at <https://github.com/JayaniLakshika/paper-nldr-vis-algorithm>. The Appendix includes many more details.

The R package `quollr`, available on CRAN and at <https://jayanilakshika.github.io/quollr/>, provides software accompanying this paper to fit the wireframe model representation, compute diagnostics, visualize the model in the data with `langevitour` and link multiple plots interactively.

8 Acknowledgments

These R packages were used for the work: `tidyverse` (Wickham et al. 2019), `Rtsne` (Krijthe 2015), `umap` (Konopka 2023), `patchwork` (Pedersen 2024), `colorspace` (Zeileis et al. 2020), `langevitour` (Harrison 2023), `conflicted` (Wickham 2023), `reticulate` (Ushey et al. 2024), `kableExtra` (Zhu 2024). These python packages were used for the work: `trimap`

(Amid & Warmuth 2019) and `pacmap` (Wang et al. 2021). The article was created with R packages `quarto` (Allaire & Dervieux 2024).

References

- 10x Genomics (2016), ‘3k PBMCs from a Healthy Donor, Universal 3’ Gene Expression dataset analysed using Cell Ranger 1.1.0’. Accessed: 2025-06-09. <https://www.10xgenomics.com/datasets/3-k-pbm-cs-from-a-healthy-donor-1-standard-1-1-0>.
- Allaire, J. & Dervieux, C. (2024), *quarto: R Interface to Quarto Markdown Publishing System*. R package version 1.4.4. <https://CRAN.R-project.org/package=quarto>.
- Amid, E. & Warmuth, M. K. (2019), ‘TriMap: Large-scale Dimensionality Reduction Using Triplets’, *arXiv preprint arXiv:1910.00204* .
- Asimov, D. (1985), ‘The Grand Tour: A Tool for Viewing Multidimensional Data’, *SIAM Journal of Scientific and Statistical Computing* **6**(1), 128–143.
- Barber, C. B., Dobkin, D. P. & Huhdanpaa, H. (1996), ‘The Quickhull Algorithm for Convex Hulls’, *ACM Trans. Math. Softw.* **22**(4), 469—483. <https://doi.org/10.1145/235815.235821>.
- Borg, I. & Groenen, P. J. F. (2005), *Modern Multidimensional Scaling*, 2nd edn, Springer, New York, NY. <https://doi.org/10.1007/0-387-28981-X>.
- Brown, T. A. (2015), *Confirmatory Factor Analysis for Applied Research*, 2nd edn, The Guilford Press, New York, NY, US.

- Carr, D. B., Littlefield, R. J., Nicholson, W. L. & Littlefield, J. S. (1987), ‘Scatterplot Matrix Techniques for Large N’, *Journal of the American Statistical Association* **82**(398), 424–436. <http://www.jstor.org/stable/2289444>.
- Carr, D., Lewin-Koh, N., Maechler, M. & Sarkar, D. (2023), *hexbin: Hexagonal Binning Routines*. R package version 1.28.3. <https://CRAN.R-project.org/package=hexbin>.
- Chari, T. & Pachter, L. (2023), ‘The specious art of single-cell genomics’, *PLoS Computational Biology* **19**, e1011288. <https://doi.org/10.1371/journal.pcbi.1011288>
- Chen, Z., Wang, C., Huang, S., Shi, Y. & Xi, R. (2024), ‘Directly Selecting Cell-type Marker Genes for Single-cell Clustering Analyses’, *Cell Reports Methods* **4**(7), 100810. <https://www.sciencedirect.com/science/article/pii/S2667237524001735>.
- Coifman, R. R., Lafon, S., Lee, A. B., Maggioni, M., Nadler, B., Warner, F. & Zucker, S. W. (2005), ‘Geometric Diffusions as a Tool for Harmonic Analysis and Structure Definition of Data: Diffusion Maps’, *Proceedings of the National Academy of Sciences of the United States of America* **102**(21), 7426–7431. <https://www.pnas.org/doi/abs/10.1073/pnas.0500334102>.
- Freedman, D. A. & Diaconis, P. (1981), ‘On the Histogram as a Density Estimator: L2 Theory’, *Probability Theory and Related Fields* **57**, 453–476. <https://doi.org/10.1007/BF01025868>.
- Gebhardt, A., Bivand, R. & Sinclair, D. (2024), *interp: Interpolation Methods*. R package version 1.1-6. <https://CRAN.R-project.org/package=interp>.
- Haque, A., Engel, J., Teichmann, S. A. & Lönnberg, T. (2017), ‘A Practical Guide to

- Single-cell RNA-sequencing for Biomedical Research and Clinical Applications’, *Genome Medicine* **9**(1), 75. <https://doi.org/10.1186/s13073-017-0467-4>.
- Harrison, P. (2023), ‘langevitour: Smooth Interactive Touring of High Dimensions, Demonstrated With scRNA-Seq Data’, *The R Journal* **15**, 206–219. <https://doi.org/10.32614/RJ-2023-046>.
- Hart, C. & Wang, E. (2022), *detourr: Portable and Performant Tour Animations*. R package version 0.1.0. <https://casperhart.github.io/detourr/>.
- Hinton, G. E. & Salakhutdinov, R. R. (2006), ‘Reducing the dimensionality of data with neural networks’, *Science* **313**(5786), 504–507.
- Irizarry, R. (2024), ‘Biologists, stop putting umap plots in your papers’, <https://simplystatistics.org/posts/2024-12-23-biologists-stop-including-umap-plots-in-your-papers/>. Blog post on Simply Statistics.
- Johnstone, I. M. & Titterton, D. M. (2009), ‘Statistical Challenges of High-Dimensional Data’, *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* **367**(1906), 4237–4253. <https://royalsocietypublishing.org/doi/abs/10.1098/rsta.2009.0159>.
- Jolliffe, I. (2011), *Principal Component Analysis*, Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 1094–1096. https://doi.org/10.1007/978-3-642-04898-2_455.
- Jöreskog, K. G. (1969), ‘A General Approach to Confirmatory Maximum Likelihood Factor Analysis’, *Psychometrika* **34**(2), 183–202. <https://doi.org/10.1007/BF02289343>.

- Konopka, T. (2023), *umap: Uniform Manifold Approximation and Projection*. R package version 0.2.10.0. <https://CRAN.R-project.org/package=umap>.
- Krijthe, J. H. (2015), *Rtsne: T-Distributed Stochastic Neighbor Embedding Using Barnes-Hut Implementation*. R package version 0.16. <https://github.com/jkrijthe/Rtsne>.
- Kruskal, J. B. (1964), ‘Nonmetric Multidimensional Scaling: A Numerical Method’, *Psychometrika* **29**(2), 115–129. <https://doi.org/10.1007/BF02289694>.
- Laa, U., Cook, D. & Lee, S. (2022), ‘Burning Sage: Reversing the Curse of Dimensionality in the Visualization of High-Dimensional Data’, *Journal of Computational and Graphical Statistics* **31**(1), 40–49. <https://doi.org/10.1080/10618600.2021.1963264>.
- Laurent, S. (2023), *cxhull: Convex Hull*. R package version 0.7.4. <https://github.com/stla/cxhull>.
- LeCun, Y., Cortes, C. & Burges, C. J. C. (1998), ‘The MNIST Database of Handwritten Digits’. Accessed: 2025-06-09. <http://yann.lecun.com/exdb/mnist/>.
- Lee, D. T. & Schachter, B. J. (1980), ‘Two Algorithms For Constructing a Delaunay Triangulation’, *International Journal of Computer & Information Sciences* **9**(3), 219–242. <https://doi.org/10.1007/BF00977785>.
- Lee, J. A., Peluffo-Ordóñez, D. H. & Verleysen, M. (2015), ‘Multi-scale similarities in stochastic neighbour embedding: Reducing dimensionality while preserving both local and global structure’, *Neurocomputing* **169**, 246–261. <https://www.sciencedirect.com/science/article/pii/S0925231215003641>

- Lee, S., Cook, D., da Silva, N., Laa, U., Wang, E., Spyrisson, N. & Zhang, H. S. (2021), ‘A Review of the State-of-the-Art on Tours for Dynamic Visualization of High-Dimensional Data’, *arXiv preprint arXiv:2104.08016* .
- Maaten, L. V. D. & Hinton, G. E. (2008), ‘Visualizing Data Using t-SNE’, *Journal of Machine Learning Research* **9**(11), 2579–2605. <https://api.semanticscholar.org/CorpusID:5855042>.
- McInnes, L., Healy, J., Saul, N. & Großberger, L. (2018), ‘UMAP: Uniform Manifold Approximation and Projection’, *Journal of Open Source Software* **3**(29), 861. <https://doi.org/10.21105/joss.00861>.
- Moon, K. R., van Dijk, D., Wang, Z., Gigante, S. A., Burkhardt, D. B., Chen, W. S., Yim, K., van den Elzen, A., Hirn, M. J., Coifman, R. R., Ivanova, N. B., Wolf, G. & Krishnaswamy, S. (2019), ‘Visualizing Structure and Transitions in High-Dimensional Biological Data’, *Nature Biotechnology* **37**, 1482–1492. <https://doi.org/10.1038/s41587-019-0336-3>.
- Pedersen, T. L. (2024), *patchwork: The Composer of Plots*. R package version 1.2.0. <https://CRAN.R-project.org/package=patchwork>.
- Saeed, N., Nam, H., Haq, M. I. U. & Muhammad Saqib, D. B. (2018), ‘A Survey on Multidimensional Scaling’, *ACM Comput. Surv.* **51**(3). <https://doi.org/10.1145/3178155>.
- Shepard, R. N. (1962), ‘The analysis of proximities: Multidimensional scaling with an unknown distance function. i.’, *Psychometrika* **27**(2), 125–140.
- Silva, V. & Tenenbaum, J. (2002), ‘Global Versus Local Methods in Non-linear Dimensionality Reduction’, *Advances in Neural Information Processing*

- Systems* **15**, 721–728. https://proceedings.neurips.cc/paper_files/paper/2002/file/5d6646aad9bcc0be55b2c82f69750387-Paper.pdf.
- Spearman, C. (1961), ‘The proof and measurement of association between two things.’
- Ushey, K., Allaire, J. & Tang, Y. (2024), *reticulate: Interface to Python*. R package version 1.38.0. <https://CRAN.R-project.org/package=reticulate>.
- van der Maaten, L. (2009), Learning a parametric embedding by preserving local structure, *in* ‘Artificial Intelligence and Statistics’, PMLR, pp. 384–391.
- Wang, Y., Huang, H., Rudin, C. & Shaposhnik, Y. (2021), ‘Understanding How Dimension Reduction Tools Work: An Empirical Approach to Deciphering t-SNE, UMAP, TriMap, and PaCMAP for Data Visualization’, *Journal of Machine Learning Research* **22**(201), 1–73. <http://jmlr.org/papers/v22/20-1061.html>.
- Wickham, H. (2023), *conflicted: An Alternative Conflict Resolution Strategy*. R package version 1.2.0. <https://CRAN.R-project.org/package=conflicted>.
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Golemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K. & Yutani, H. (2019), ‘Welcome to the Tidyverse’, *Journal of Open Source Software* **4**(43), 1686. <https://doi.org/10.21105/joss.01686>.
- Wickham, H., Cook, D. & Hofmann, H. (2015), ‘Visualizing Statistical Models: Removing the Blindfold’, *Statistical Analysis and Data Mining: The ASA Data Science Journal* **8**(4), 203–225. <https://onlinelibrary.wiley.com/doi/abs/10.1002/sam.11271>.

- Wickham, H., Cook, D., Hofmann, H. & Buja, A. (2011), ‘tourr: An R Package For Exploring Multivariate Data With Projections’, *Journal of Statistical Software* **40**(2), 1–18. <http://www.jstatsoft.org/v40/i02/>.
- Zeileis, A., Fisher, J. C., Hornik, K., Ihaka, R., McWhite, C. D., Murrell, P., Stauffer, R. & Wilke, C. O. (2020), ‘colorspace: A Toolbox for Manipulating and Assessing Colors and Palettes’, *Journal of Statistical Software* **96**(1), 1–49. <https://www.jstatsoft.org/index.php/jss/article/view/v096i01>.
- Zhu, H. (2024), *kableExtra: Construct Complex Table with kable and Pipe Syntax*. R package version 1.4.0. <https://CRAN.R-project.org/package=kableExtra>.